DSA 5103

Project Proposal

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Project Title: Loan Approval Prediction

Description of the problem context:

Loan approval is the process by which financial institutions, such as banks or credit unions, evaluate and decide whether to grant a loan to an individual or entity. This decision is based on an assessment of the applicant's financial health, credit history, ability to repay the loan, and numerous other factors. The applicant fills out a form that provides information which is then used by financial institutions to arrive at a decision regarding the loan approval.

The dataset contains features obtained from individual applicant details, financial metric and other loan factors. The objective here is to predict for a given set of input features collected during the application process whether the loan will be approved or not. This dataset is collected from Kaggle, the source of the data is not known.

Type of the problem: Supervised learning. This is a classification problem, more specifically binary classification. The training data can be assigned to two different classes based on whether the loan is approved or not. A model can be used to assign new data to these classes. Methods including logistic regression, random forest, k-NN and support vector machines can be investigated.

Initial thoughts on the technique that might be used: Preprocess the dataset and explore techniques for handling imbalanced classes (for e.g., more approvals than denials), handle missing values using imputation techniques (for e.g. k-NN classifier), feature transformation. Conduct exploratory data analysis for better understanding of data. Construct the model to predict which loans will be approved and which will be rejected. Use cross-validation for data resampling. Validate the model using evaluation metrics that account for class imbalances like F1 score and ROC-AUC.

Hyperlink of the problem description/source of dataset:
https://www.kaggle.com/datasets/itshappy/ps4e9-original-data-loan-approval-prediction